Whose Side Is the Bidding Assistant On?

AI alignment, advice following, and equilibrium selection in a first-price auction

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KEY MESSAGE

The model has two Nash equilibria. Buyer advice favors Low–Low. Seller advice makes High a best reply when beliefs exceed 2/3. Human equilibrium selection remains untested.

01 Question & model

Research question & verified gap

How does disclosed sponsor advice affect equilibrium selection and surplus?

Malenko & Tsoy (2019) study advised buyers. Logg et al. (2019) study algorithm appreciation. Our bounded gap concerns disclosed sponsor goals in a two-buyer coordination auction.

Strategic model

Two buyers bid 2 or 5 for value 6 with budget 5, in units of 10 RMB. Highest bid wins and pays its bid.
A coin flip breaks ties.
Both know the same sponsor and advice. Complete information, simultaneous moves, Nash equilibrium.

Buyer 1 \ Buyer 2	Low (2)	High (5)
Low (2)	7, 7 Nash	5, 6
High (5)	6, 5	5.5, 5.5 Nash

Belief threshold **High is strictly better when P(other bids High) > 2/3.**

At p = 2/3 both bids tie. With assumed b = 0.25, High is better for t > 5/9.

Three complementary lenses

Game theory — advice selects Low–Low or High–High.
Social choice — total welfare stays 16; advice redistributes surplus.
Mechanism design — under second price, High is dominant and High–High is unique.

Auction comparison

First price: coordination problem; sponsor-aligned advice matters.
Second price: High is strictly dominant; seller advice adds nothing.
Boundary: known common value and two bid levels—this is coordination, not private-information bid shading.

02 Design & evidence

Research design

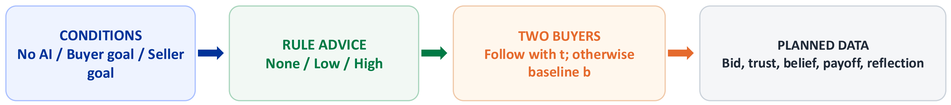


Figure 1. Deterministic sponsor advice. No trained or live LLM has been tested.

Main result: alignment changes revenue



Figure 2. Exact expectations with assumed b = 0.25. As t rises, buyer advice lowers revenue and seller advice raises it. No-AI revenue stays 3.3125. Curves meet at t = 0. The 5/9 threshold concerns best replies, not a revenue crossing. Welfare stays 16.

ACTUAL COMPUTATION

✓ Two Nash equilibria
✓ p* = 2/3; t* = 0.556
✓ Seed 42; N = 10,000 check
✓ 8 tests pass
GitHub tag v1-review-final
commit e29d9cb

PLANNED BEHAVIOR

Exploratory rounds: no AI, buyer goal, seller goal. Add P(other bids High) before bidding, alongside trust (1–7).
Report valid n, High shares and confidence intervals. No observations supplied.

03 Impact & dialogue

Practical impact: campus resale

Case: an AI bidding assistant on a campus second-hand platform.
Decision informed: whether to deploy bid suggestions and how to disclose whose objective the tool serves.
Distribution: buyer-aligned advice benefits buyers; seller-aligned advice raises seller revenue without increasing welfare.

Correction route

1 Mandatory sponsor label
2 Access to buyer- and seller-side advice
3 Queryable audit log

Interdisciplinary evidence

Trust and strategic beliefs can both explain following. Record each before the bid.

Low trust with belief above 2/3 is consistent with strategic following. High trust with belief below 2/3 motivates a direct-trust explanation.

These associations cannot establish causality. Reflections help interpret them.

Limitations & next test

Assumptions: common value, two bids, one round, independent following. t is swept over [0,1], not measured trust. b = 0.25 is assumed.

Next: estimate b with a confidence interval and recompute t*. Report three-condition bids and uncertainty. Put no AI first in a confirmatory test. Live-LLM validation remains planned.

QUESTION FOR REVIEWERS

Does seller-aligned advice change bids because buyers trust the AI—or because they expect the other buyer to follow?

SEP 28 peer review · OCT 5 response and revision

Acknowledgments

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Selected references

Harsanyi & Selten (1988), A General Theory of Equilibrium Selection in Games. Malenko & Tsoy (2019), AER, doi:10.1257/aer.20170334. Dietvorst et al. (2015), JEP:G, doi:10.1037/xge0000033. Logg et al. (2019), OBHDP, doi:10.1016/j.obhdp.2018.12.005. Greiner et al. (2025), Management Science, doi:10.1287/mnsc.2022.02777.

Open materials

GitHub github.com/dku-comsci-econ206-Autumn2026/ps2-ai-advice-auction · tag v1-review-final

Hugging Face huggingface.co/spaces/dku-comsci-econ206-2026/ps2-ai-advice-auction

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